

# State-Dependent Government Spending Multipliers in the Western Balkans and South-Eastern Europe

A Regional Panel Study Using State-Dependent Local Projections  
(Owyang–Ramey–Zubairy 2013; Ramey–Zubairy 2018)

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## Abstract

This paper estimates state-dependent government spending multipliers for eight South-Eastern European economies—Bulgaria, Croatia, Greece, Montenegro, North Macedonia, Romania, Serbia, and Slovenia—using the local projection instrumental variable (LP-IV) framework of Owyang et al. (2013), scaled following Ramey and Zubairy (2018). Business-cycle states are identified by whether the unemployment rate exceeds its Hodrick–Prescott trend. Pooled panel estimates yield no statistically significant difference between slack and expansion multipliers at any horizon up to twelve quarters; at the twelve-quarter horizon the slack multiplier is  $\hat{\beta}_{\text{slack}} = 1.19$  and the expansion multiplier is  $\hat{\beta}_{\text{exp}} = 1.30$ , both meaningfully above unity. Country-level results reveal substantial heterogeneity: Slovenia and Croatia display large, statistically significant slack multipliers (up to 5.0 and 4.3 respectively), consistent with the Keynesian demand-slack hypothesis, while Romania and Serbia exhibit multipliers below unity or negative during recessions. Robustness checks using percentile-based slack thresholds show that the median-split definition ( $\mathcal{P}_{50}$ ) produces significant state dependence at medium horizons ( $p < 0.001$ ), whereas the Hodrick–Prescott baseline and the  $\mathcal{P}_{25}$  and  $\mathcal{P}_{75}$  thresholds do not. These findings suggest that institutional capacity, exchange-rate regime, and fiscal credibility—rather than the cyclical position per se—are the primary drivers of fiscal transmission heterogeneity in the region.

**Keywords:** fiscal multipliers, state dependence, local projections, instrumental variables, Western Balkans, South-Eastern Europe, Eurostat.

**JEL Classification:** E62, H50, C23, C26.

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# 1 Introduction

A central question in macroeconomics is whether the output effect of a government spending increase depends on the state of the business cycle. The theoretical presumption—rooted in models with nominal rigidities and demand-driven recessions—is that multipliers should be larger when the economy operates below potential, because idle resources can be put to productive use without displacing private spending (Keynes, 1936; Woodford, 2011). At the zero lower bound on nominal interest rates, where monetary accommodation is automatic, this effect may be especially pronounced.

Empirical evidence on state-dependent fiscal multipliers, however, remains mixed. Auerbach and Gorodnichenko (2012) and Owyang et al. (2013) find significantly larger multipliers during recessions for the United States, using smooth-transition and threshold methods respectively. Ramey and Zubairy (2018), applying a more demanding instrumental variable strategy to a longer US historical sample, find little evidence of state dependence. The international evidence is similarly inconclusive: Ilzetzi et al. (2013) document that multipliers in developing economies and fixed exchange-rate regimes tend to be smaller, but do not focus on business-cycle asymmetries.

This paper contributes to the literature in three ways. First, we provide the first systematic estimates of state-dependent fiscal multipliers for the Western Balkans and South-Eastern Europe—a region largely absent from the empirical fiscal multiplier literature despite its policy relevance, given the bloc’s proximity to the European Union, its heterogeneous institutional arrangements, and the frequency of fiscal consolidation programmes in the post-2008 period. Second, we apply the LP-IV methodology of Owyang et al. (2013) with the scaling convention of Ramey and Zubairy (2018), now standard in the literature, to a quarterly panel of eight countries spanning up to 28 years. Third, we conduct a systematic robustness analysis across four definitions of economic slack, shedding light on the sensitivity of state-dependent conclusions to the measurement of the cycle.

Our main findings are as follows. At the pooled panel level, we find no statistically significant state dependence in fiscal multipliers. Both slack and expansion multipliers are positive and exceed unity at medium-run horizons, suggesting that government spending is on average effective in the region. At the country level, however, results are heterogeneous: Slovenia and Croatia exhibit large, statistically significant slack multipliers consistent with the Keynesian hypothesis, while Romania and Serbia display weak or negative multipliers during downturns.

The remainder of the paper is organised as follows. Section 2 describes the methodology and identification strategy. Section 3 presents the data and preliminary diagnostics. Section 4 reports the main estimation results. Section 5 discusses the findings and Section 6 concludes.

## 2 Methodology

### 2.1 Local Projection Estimator

We estimate state-dependent impulse responses using the local projection (LP) framework of Jordà (2005) as extended to state dependence by Owyang et al. (2013). Let  $Y_t$  denote real GDP and  $G_t$  real government consumption expenditure, both seasonally adjusted. Define the Ramey–Zubairy scaled variables

$$y_t^s = \frac{Y_t - Y_{t-1}}{Y_{t-1}} \times 100, \quad g_t^s = \frac{G_t - G_{t-1}}{Y_{t-1}} \times 100, \quad (1)$$

which express changes in output and spending as percentages of lagged GDP. At each projection horizon  $h = 0, 1, \dots, H$ , the cumulative outcome and spending variables are

$$\mathcal{Y}_{t,h} = \sum_{s=0}^h y_{t+s}^s, \quad \mathcal{G}_{t,h} = \sum_{s=0}^h g_{t+s}^s. \quad (2)$$

### 2.2 State Classification

The binary state variable is defined using the deviation of the unemployment rate from its Hodrick–Prescott trend (Hodrick and Prescott, 1997), computed country-by-country with smoothing parameter  $\lambda = 1,600$  appropriate for quarterly data:

$$S_t = \begin{cases} 1 & \text{(slack) if } u_t > u_t^{\text{HP}}, \\ 0 & \text{(expansion) otherwise.} \end{cases} \quad (3)$$

Following Owyang et al. (2013), the state is lagged one period so that  $S_{t-1}$  is determined before the government spending shock occurs, avoiding contemporaneous endogeneity of the state classification.

**Figure 1: Unemployment Rate and HP Trend – Balkan Region**

Shaded: quarters identified as 'slack' (unemp &gt; HP trend, lagged 1Q). Source: Eurostat/World Bank

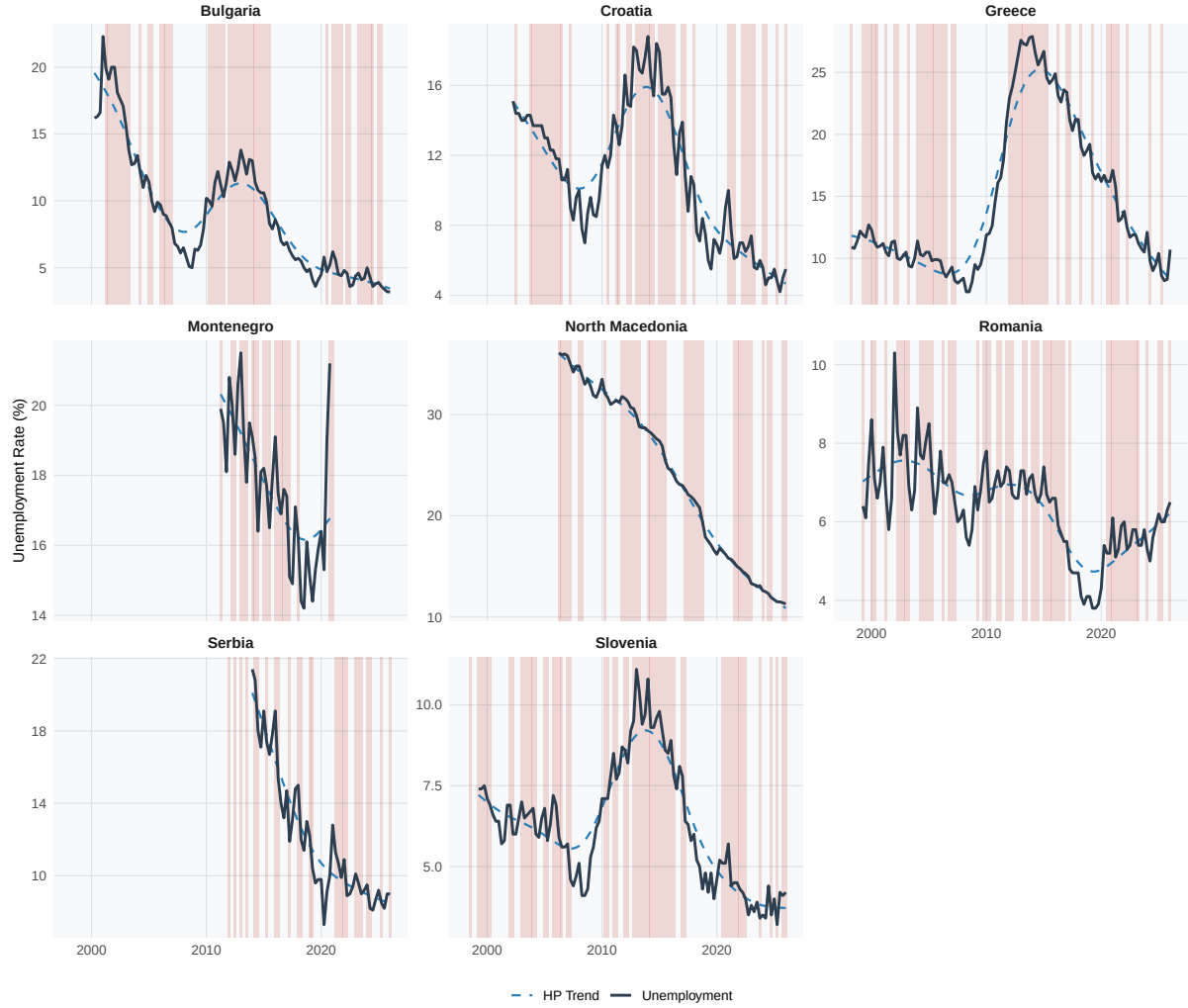


Figure 1: Unemployment Rate and HP Trend - Balkan Region

## 2.3 Instrumental Variable Estimation

Government spending is endogenous because policymakers may respond to output conditions within the quarter. Following Blanchard and Perotti (2002), we instrument for cumulative spending  $\mathcal{G}_{t,h}$  with the contemporaneous spending change  $g_t^s$ , which is predetermined within the quarter under the Blanchard–Perotti identification assumption. The interacted 2SLS specification is:

$$\mathcal{Y}_{t,h} = \alpha_{\text{slack}} \mathcal{G}_{t,h}^{\text{slack}} + \alpha_{\text{exp}} \mathcal{G}_{t,h}^{\text{exp}} + \gamma' \mathbf{X}_{t-1} + \mu_i + \varepsilon_{t,h}, \quad (4)$$

where  $\mathcal{G}_{t,h}^j = S_{t-1}^j \cdot \mathcal{G}_{t,h}$  for  $j \in \{\text{slack}, \text{exp}\}$ , instrumented by  $z_t^j = S_{t-1}^j \cdot g_t^s$ . The vector  $\mathbf{X}_{t-1}$  contains two lags each of  $y^s$  and  $g^s$ ;  $\mu_i$  are country fixed effects in the panel specification; and  $\varepsilon_{t,h}$  is the error term. Standard errors are clustered by country in the pooled panel and Newey–West HAC with bandwidth  $h + 1$  in the country-by-country specifications (Newey and West, 1987).

The cumulative multipliers  $\hat{\beta}_{\text{slack}}^h$  and  $\hat{\beta}_{\text{exp}}^h$  are the 2SLS estimates of  $\alpha_{\text{slack}}$  and  $\alpha_{\text{exp}}$  in equation (4), directly interpretable as the local-currency increase in cumulative GDP per unit increase in cumulative government spending, conditional on the initial state of the economy.

### 3 Data and Preliminary Analysis

#### 3.1 Sample Coverage

The empirical analysis draws on seasonally adjusted quarterly national accounts data from Eurostat (`namq_10_gdp`) and quarterly labour force survey data (`lfsq_organ`) for eleven Balkan and South-Eastern European economies (Eurostat, 2013). Following the data quality screening described above, Bosnia and Herzegovina is excluded due to an insufficient number of usable quarters ( $T = 19 < 24$ ). The final estimation sample covers eight countries: Bulgaria (BG), Croatia (HR), Greece (EL), Montenegro (ME), North Macedonia (MK), Romania (RO), Serbia (RS), and Slovenia (SI), with time series ranging from 1998:Q2 for Greece, Romania, and Slovenia to 2026:Q1 for Bulgaria, Greece, North Macedonia, Romania, and Slovenia. Table 1 summarises coverage details.

Table 1: Data Coverage by Country

| Country         | Code | Start   | End     | $T$ | % Slack |
|-----------------|------|---------|---------|-----|---------|
| Bulgaria        | BG   | 2000:Q2 | 2026:Q1 | 104 | 51.0    |
| Croatia         | HR   | 2002:Q2 | 2026:Q1 | 96  | 49.0    |
| Greece          | EL   | 1998:Q2 | 2026:Q1 | 112 | 45.5    |
| Montenegro      | ME   | 2011:Q2 | 2021:Q1 | 40  | 52.5    |
| North Macedonia | MK   | 2006:Q2 | 2026:Q1 | 80  | 51.2    |
| Romania         | RO   | 1998:Q3 | 2026:Q1 | 109 | 52.3    |
| Serbia          | RS   | 2010:Q3 | 2026:Q1 | 56  | 46.4    |
| Slovenia        | SI   | 1998:Q3 | 2026:Q1 | 109 | 54.1    |

*Notes:*  $T$  denotes usable quarters after constructing lagged controls. “% Slack” is the fraction of quarters in the slack state (lagged unemployment above HP trend). Bosnia and Herzegovina ( $T = 19$ ) is excluded. Source: Eurostat `namq_10_gdp` and `lfsq_organ`.

Across the estimation sample, slack quarters account for roughly half of all observations in each country, with the share ranging from 45.5% in Greece to 54.1% in Slovenia. This near-even split is a desirable feature of the identification strategy, ensuring that multiplier estimates in both regimes are anchored by a sufficiently large number of observations.

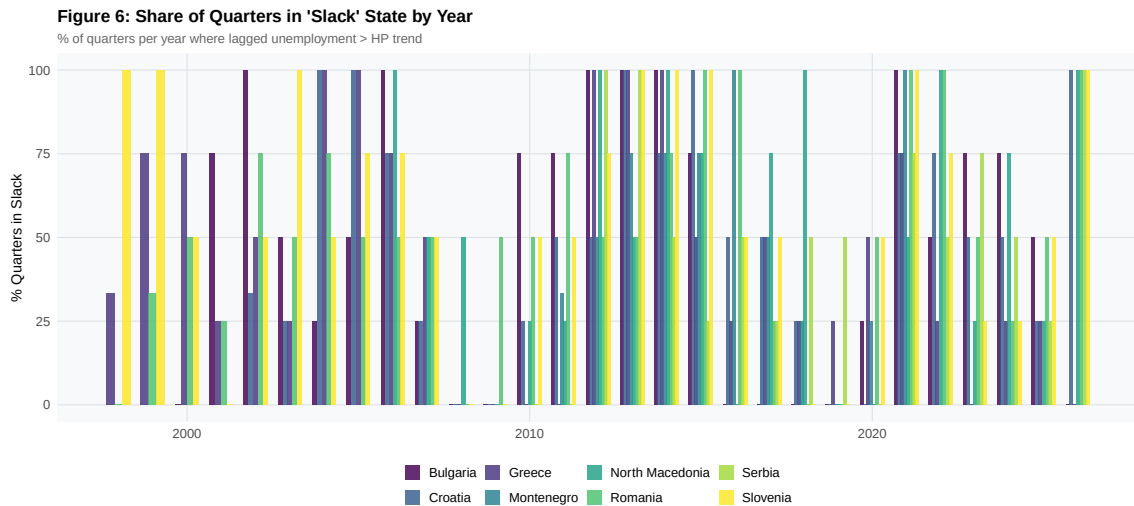


Figure 2: Share of Quarters in 'Slack' State by Year

### 3.2 Descriptive Statistics

Table 2 reports means and standard deviations of the key variables by country and business-cycle state. Several patterns emerge. First, output growth is uniformly higher during slack periods in six of the eight countries, likely reflecting mean-reverting behaviour of GDP following contractionary episodes rather than a causal effect of fiscal policy. Second, government expenditure growth is broadly procyclical: positive and slightly higher during expansion in Bulgaria, Croatia, and Slovenia, while Montenegro and Romania exhibit negative or near-zero government expenditure growth during slack, consistent with fiscal consolidation pressures coinciding with downturns. Third, the standard deviation of GDP growth is substantially larger during expansions—notably Montenegro ( $\sigma = 4.90$  vs. 3.84), Croatia (2.91 vs. 1.33), and Greece (2.53 vs. 1.42)—suggesting that the slack state is associated with more stable, though depressed, economic conditions.



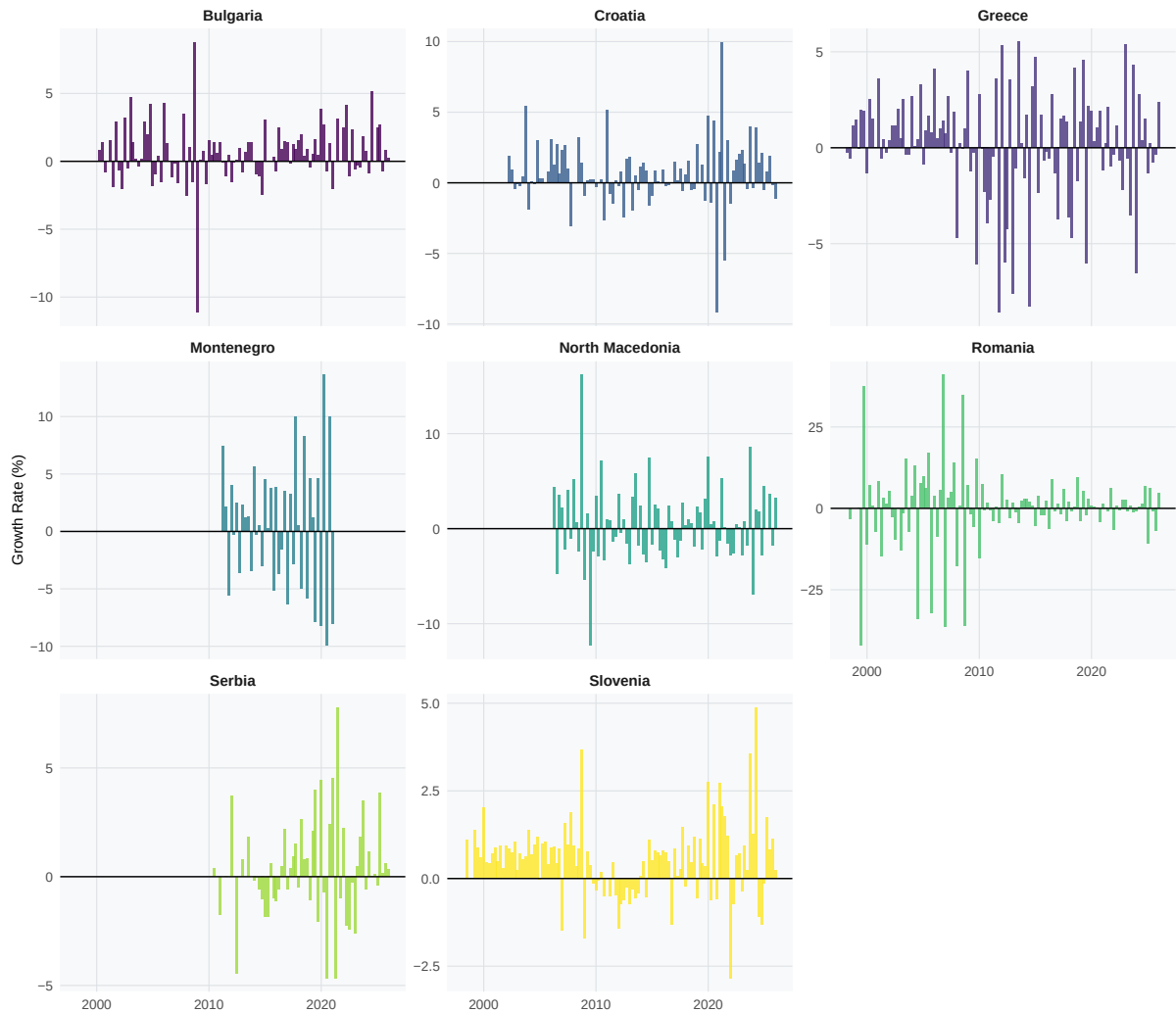
Table 2: Descriptive Statistics by Country and Business-Cycle State

| Country      | State     | GDP growth (%) |      | Gov. exp. growth (%) |       | Unemp. (%) | Obs.     |          |
|--------------|-----------|----------------|------|----------------------|-------|------------|----------|----------|
|              |           | Mean           | S.D. | Mean                 | S.D.  | Mean       | <i>N</i> | <i>T</i> |
| Bulgaria     | Expansion | 0.56           | 1.46 | 0.62                 | 2.46  | 7.56       | 51       | 104      |
|              | Slack     | 1.02           | 0.92 | 0.60                 | 1.86  | 10.30      | 53       |          |
| Croatia      | Expansion | 0.32           | 2.91 | 0.44                 | 2.26  | 9.59       | 49       | 96       |
|              | Slack     | 0.81           | 1.33 | 0.84                 | 2.24  | 12.12      | 47       |          |
| Greece       | Expansion | −0.24          | 2.53 | −0.10                | 2.72  | 13.33      | 61       | 112      |
|              | Slack     | 0.76           | 1.42 | 0.52                 | 2.95  | 16.24      | 51       |          |
| Montenegro   | Expansion | −0.94          | 4.90 | −0.62                | 6.68  | 16.98      | 19       | 40       |
|              | Slack     | 1.71           | 3.84 | 1.47                 | 4.21  | 18.22      | 21       |          |
| N. Macedonia | Expansion | 0.63           | 3.73 | 0.92                 | 4.69  | 23.26      | 39       | 80       |
|              | Slack     | 0.63           | 1.04 | −0.07                | 3.00  | 24.31      | 41       |          |
| Romania      | Expansion | 0.59           | 3.00 | 0.69                 | 12.86 | 6.05       | 52       | 109      |
|              | Slack     | 1.02           | 1.96 | −0.65                | 10.69 | 6.65       | 57       |          |
| Serbia       | Expansion | 0.54           | 2.71 | 0.35                 | 2.05  | 12.06      | 30       | 56       |
|              | Slack     | 0.75           | 1.09 | 0.30                 | 2.63  | 12.17      | 26       |          |
| Slovenia     | Expansion | 0.17           | 1.99 | 0.60                 | 1.11  | 5.51       | 50       | 109      |
|              | Slack     | 0.95           | 1.79 | 0.50                 | 1.06  | 6.91       | 59       |          |

*Notes:* GDP growth and government expenditure growth are quarter-on-quarter log-differences multiplied by 100, scaled by lagged GDP following Ramey and Zubairy (2018). “Expansion” (`slack_lag1` = 0) and “Slack” (`slack_lag1` = 1) denote the lagged business-cycle state. Unemployment is in percentage points.

**Figure 2: Real Government Expenditure Growth (QoQ, %)**

Seasonally adjusted chain-linked volumes. Source: Eurostat namq\_10\_gdp

**Figure 3: Real Government Expenditure Growth (QoQ, %)**

### 3.3 Unit Root Tests

Before proceeding to estimation, we test for unit roots in the log levels and growth rates of GDP and government expenditure using the Augmented Dickey–Fuller (ADF) test (Dickey and Fuller, 1979). The results, reported in Table 3, are consistent with standard findings: log levels ( $\ln Y_t$ ,  $\ln G_t$ ) are non-stationary in all eight countries, while growth rates ( $\Delta y_t$ ,  $\Delta g_t$ ) are stationary at the 5% level in the majority of cases. The one notable exception is Greek GDP growth ( $p = 0.063$ ), which marginally fails to reject the unit root null, likely reflecting the unusual trajectory of Greek output during and after the 2010–2018 sovereign debt crisis. Given that the local projection estimator is applied to

the Ramey–Zubairy scaled variables—which are in growth-rate form relative to lagged GDP—stationarity of the first-differenced series is the relevant condition, and this holds broadly across the sample.

## 4 Estimation Results

### 4.1 Instrument Validity: First-Stage Diagnostics

Table 4 reports the first-stage  $F$ -statistics at each projection horizon for the pooled panel. With the exception of  $h = 0$ —where the instrument is numerically identical to the endogenous variable because cumulative spending collapses to the contemporaneous shock— $F$ -statistics range from 18.6 to 37.6, comfortably above the Staiger and Stock (1997) rule-of-thumb threshold of 10. Weak-instrument bias is therefore not a concern for the 2SLS estimates at horizons  $h \geq 1$ .

### 4.2 Pooled Panel: Impulse Responses and Wald Tests

Table 5 presents the pooled panel impulse responses and the Wald tests of equality between slack and expansion multipliers at each horizon. The null hypothesis  $H_0: \hat{\beta}_{\text{slack}} = \hat{\beta}_{\text{exp}}$  is never rejected:  $p$ -values range from 0.21 to 0.98 across all horizons.

Despite the absence of statistically significant state dependence, the point estimates exhibit an economically interesting pattern. At short horizons ( $h \leq 3$ ) the slack multiplier exceeds the expansion multiplier, consistent with the theoretical prediction. At horizon 4 the ordering reverses ( $\hat{\beta}_{\text{slack}} = 1.57 < \hat{\beta}_{\text{exp}} = 2.47$ ), but the difference is insignificant ( $p = 0.306$ ). At medium-to-long horizons both estimates converge to the range 1.0–1.6, consistent with a modest but persistent fiscal transmission mechanism across the region.

Table 3: Augmented Dickey–Fuller Unit Root Tests

| Country      | Variable   | ADF statistic | <i>p</i> -value | Stationary |
|--------------|------------|---------------|-----------------|------------|
| Bulgaria     | $\ln Y$    | −2.641        | 0.311           | No         |
|              | $\ln G$    | −0.244        | 0.990           | No         |
|              | $\Delta y$ | −4.670        | 0.010           | Yes        |
|              | $\Delta g$ | −5.492        | 0.010           | Yes        |
| Croatia      | $\ln Y$    | −1.279        | 0.875           | No         |
|              | $\ln G$    | −0.599        | 0.976           | No         |
|              | $\Delta y$ | −4.157        | 0.010           | Yes        |
|              | $\Delta g$ | −3.920        | 0.016           | Yes        |
| Greece       | $\ln Y$    | −1.939        | 0.602           | No         |
|              | $\ln G$    | −2.187        | 0.499           | No         |
|              | $\Delta y$ | −3.372        | 0.063           | No         |
|              | $\Delta g$ | −4.310        | 0.010           | Yes        |
| Montenegro   | $\ln Y$    | −2.573        | 0.348           | No         |
|              | $\ln G$    | −3.008        | 0.178           | No         |
|              | $\Delta y$ | −3.151        | 0.122           | No         |
|              | $\Delta g$ | −4.267        | 0.010           | Yes        |
| N. Macedonia | $\ln Y$    | −3.323        | 0.074           | No         |
|              | $\ln G$    | −3.011        | 0.162           | No         |
|              | $\Delta y$ | −5.462        | 0.010           | Yes        |
|              | $\Delta g$ | −4.105        | 0.010           | Yes        |
| Romania      | $\ln Y$    | −1.755        | 0.678           | No         |
|              | $\ln G$    | −2.280        | 0.461           | No         |
|              | $\Delta y$ | −4.409        | 0.010           | Yes        |
|              | $\Delta g$ | −7.933        | 0.010           | Yes        |
| Serbia       | $\ln Y$    | −2.928        | 0.200           | No         |
|              | $\ln G$    | −2.946        | 0.193           | No         |
|              | $\Delta y$ | −4.850        | 0.010           | Yes        |
|              | $\Delta g$ | −3.655        | 0.037           | Yes        |
| Slovenia     | $\ln Y$    | −2.298        | 0.453           | No         |
|              | $\ln G$    | −1.285        | 0.873           | No         |
|              | $\Delta y$ | −4.336        | 0.010           | Yes        |
|              | $\Delta g$ | −3.644        | 0.032           | Yes        |

*Notes:* ADF test with automatic lag selection (Schwert criterion). *p*-values are bounded at 0.01 and 0.10. Null hypothesis: unit root. Rejection at the 5% level denoted “Yes”.

Table 4: First-Stage  $F$ -Statistics — Pooled Panel

| Horizon $h$ | $F$ -statistic        | Weak IV ( $F < 10$ ) |
|-------------|-----------------------|----------------------|
| 0           | $\gg 10^{30} \dagger$ | No                   |
| 1           | 27.63                 | No                   |
| 2           | 25.56                 | No                   |
| 3           | 27.77                 | No                   |
| 4           | 21.35                 | No                   |
| 5           | 31.20                 | No                   |
| 6           | 19.44                 | No                   |
| 7           | 37.58                 | No                   |
| 8           | 22.78                 | No                   |
| 9           | 18.61                 | No                   |
| 10          | 26.35                 | No                   |
| 11          | 24.57                 | No                   |
| 12          | 27.48                 | No                   |

$\dagger$ At  $h = 0$  the instrument equals the endogenous regressor exactly, producing a degenerate first stage. Retained for completeness only. For  $h \geq 1$  all  $F$ -statistics exceed 10, ruling out weak-instrument bias.

Table 5: Pooled Panel LP-IV Estimates and Wald Tests ( $H_0: \hat{\beta}_{\text{slack}}^h = \hat{\beta}_{\text{exp}}^h$ )

| Horizon $h$ | $\hat{\beta}_{\text{slack}}$ | $\hat{\beta}_{\text{exp}}$ | Difference | $p$ -value | Sig. |
|-------------|------------------------------|----------------------------|------------|------------|------|
| 0           | 0.259                        | 0.037                      | 0.222      | 0.525      |      |
| 1           | 0.677                        | 0.242                      | 0.435      | 0.748      |      |
| 2           | 0.917                        | 0.641                      | 0.276      | 0.614      |      |
| 3           | 1.317                        | 0.832                      | 0.485      | 0.214      |      |
| 4           | 1.573                        | 2.472                      | -0.899     | 0.306      |      |
| 5           | 0.725                        | 0.755                      | -0.030     | 0.961      |      |
| 6           | 1.009                        | 0.864                      | 0.145      | 0.821      |      |
| 7           | 0.827                        | 1.125                      | -0.298     | 0.555      |      |
| 8           | 1.405                        | 1.026                      | 0.379      | 0.472      |      |
| 9           | 1.117                        | 0.935                      | 0.182      | 0.799      |      |
| 10          | 0.712                        | 0.698                      | 0.014      | 0.983      |      |
| 11          | 1.104                        | 0.977                      | 0.127      | 0.846      |      |
| 12          | 1.190                        | 1.296                      | -0.106     | 0.794      |      |

Notes:  $\hat{\beta}_{\text{slack}}$  and  $\hat{\beta}_{\text{exp}}$  are 2SLS estimates of the cumulative multiplier at horizon  $h$ , scaled following Ramey and Zubairy (2018). Standard errors are clustered by country. Country fixed effects included. Significance codes: \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.10$ .

**Figure 3: State-Dependent IRFs – Pooled Balkan Panel**

Cumulative GDP response (% of lagged GDP) to 1pp spending shock  
90% & 68% CI. Clustered SE by country. Country FE included.

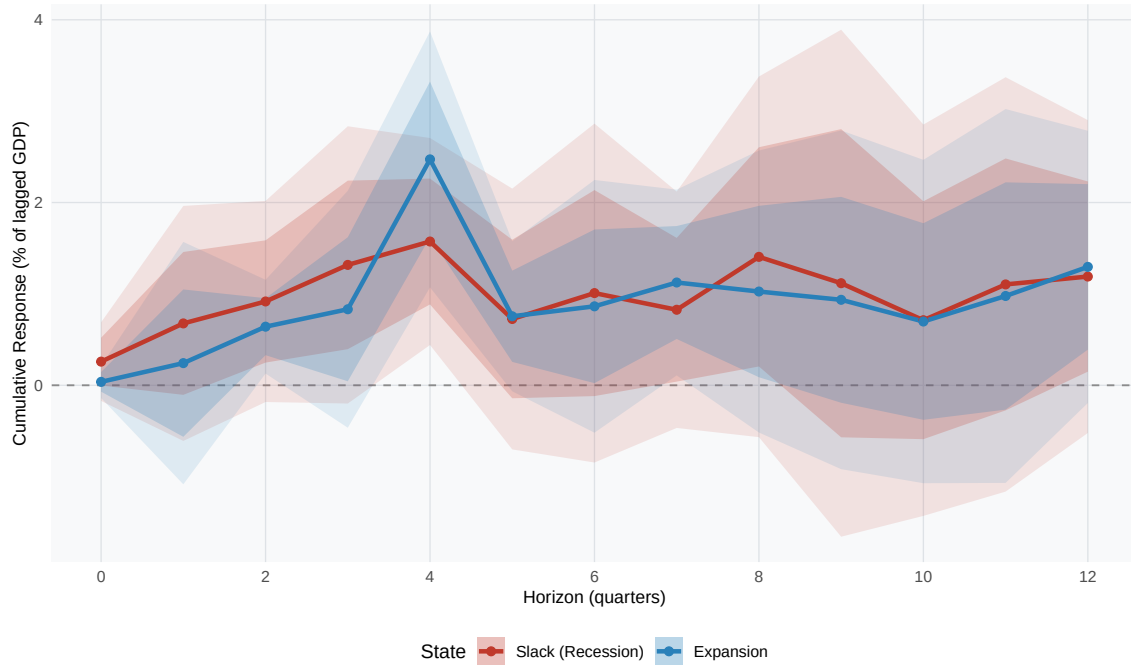


Figure 4: State-Dependent IRFs — Pooled Balkan Panel

### 4.3 Country-Level Results

Table 6 collects the cumulative multipliers at the 4-, 8-, and 12-quarter horizons for each country and the pooled panel.

Table 6: Cumulative Government Spending Multipliers by Country and Horizon

| Group          | $h = 4$ |        | $h = 8$ |        | $h = 12$ |        | Note |
|----------------|---------|--------|---------|--------|----------|--------|------|
|                | Slack   | Exp.   | Slack   | Exp.   | Slack    | Exp.   |      |
| Panel (pooled) | 1.573   | 2.472  | 1.405   | 1.026  | 1.190    | 1.296  |      |
| Bulgaria       | 1.084   | 1.006  | 2.784   | 1.641  | 2.321    | 1.826  |      |
| Croatia        | 3.774   | -1.659 | 4.342   | -0.381 | 2.309    | -1.356 | **   |
| Greece         | 2.936   | 5.782  | 4.432   | 4.485  | 3.104    | 4.513  |      |
| Montenegro     | 2.566   | 5.441  | 1.657   | -3.877 | 0.578    | 1.117  |      |
| N. Macedonia   | 1.082   | -0.103 | 0.545   | -0.550 | 3.485    | 1.604  |      |
| Romania        | 0.600   | 2.229  | -0.332  | -0.220 | -0.535   | 0.006  |      |
| Serbia         | -2.891  | 0.217  | -2.346  | 0.191  | -3.766   | -1.365 |      |
| Slovenia       | 4.143   | -1.993 | 4.596   | -0.236 | 4.971    | 1.665  | **   |

*Notes:* Estimated via 2SLS local projections with Ramey–Zubairy scaling. Standard errors are Newey–West HAC with bandwidth  $h + 1$  (country-level) or country-clustered (panel). \*\* indicates that  $H_0: \hat{\beta}_{\text{slack}} = \hat{\beta}_{\text{exp}}$  is rejected at the 5% level at least one horizon ( $h = 8$  for Croatia,  $p = 0.021$ ;  $h = 4$  for Slovenia,  $p = 0.003$ , and  $h = 8$ ,  $p = 0.031$ ).

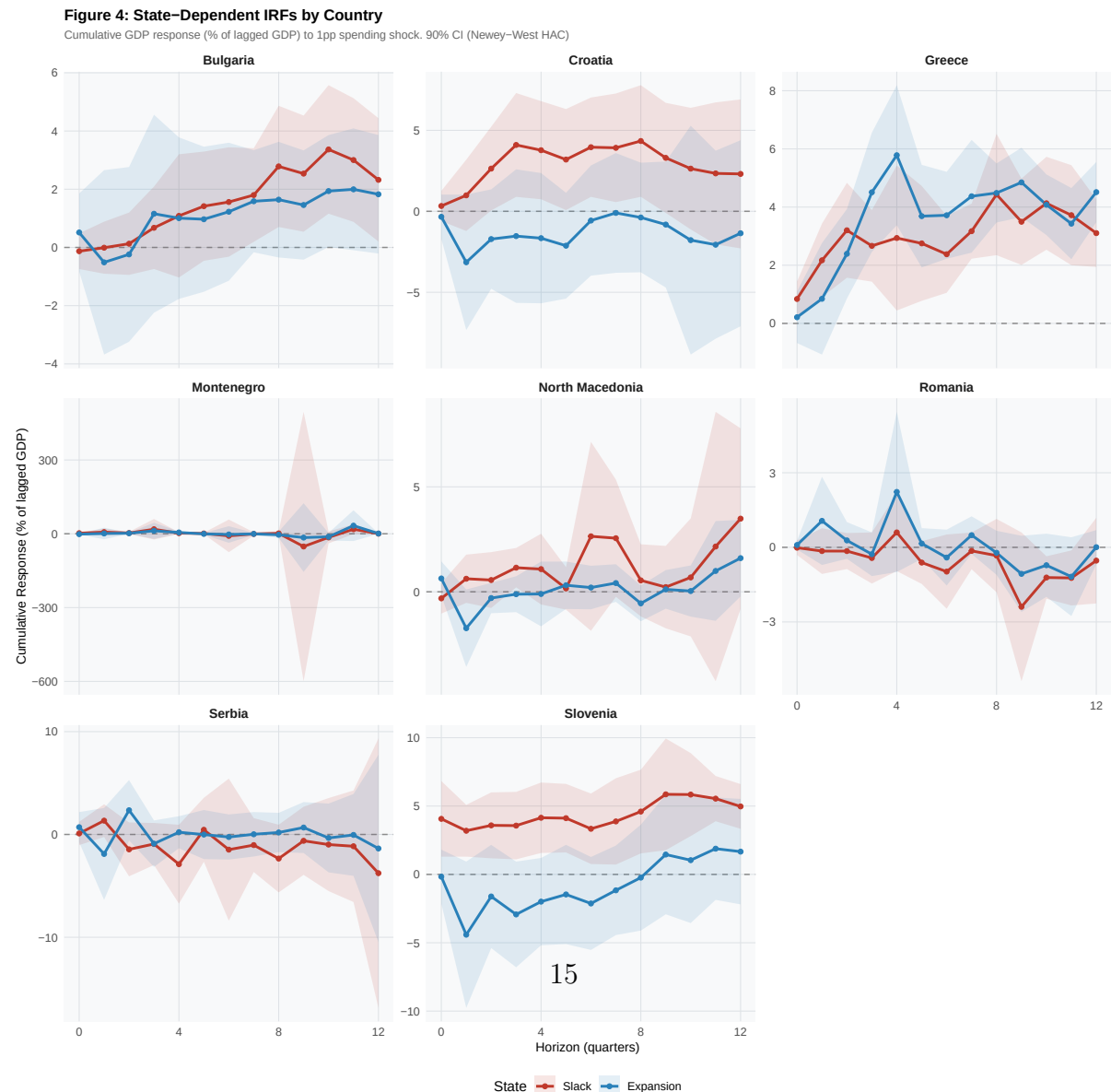


Figure 5: State-Dependent IRFs by Country

The country-level results reveal substantial cross-country heterogeneity. We organise the discussion around three groups.

**Countries with large, significant slack multipliers.** Slovenia exhibits the most pronounced state dependence in the sample. The slack multiplier rises from 4.14 at four quarters to 4.97 at twelve quarters, while the expansion multiplier is negative at short horizons ( $-1.99$  at  $h = 4$ ) and only marginally positive at longer ones. The Wald test rejects equality at  $h = 4$  ( $p = 0.003$ ) and  $h = 8$  ( $p = 0.031$ ), making Slovenia the clearest case of Keynesian state dependence in the sample. Croatia likewise displays large positive slack multipliers (3.77, 4.34, 2.31 at horizons 4, 8, and 12) and significantly negative expansion multipliers, with equality rejected at  $h = 8$  ( $p = 0.021$ ) and marginally at  $h = 4$  ( $p = 0.071$ ) and  $h = 12$  ( $p = 0.068$ ). Both countries are EU member states and euro-area members (Slovenia since 2007, Croatia since 2023), and may benefit from credible fiscal frameworks and automatic monetary accommodation that amplify the demand channel of government spending during recessions.

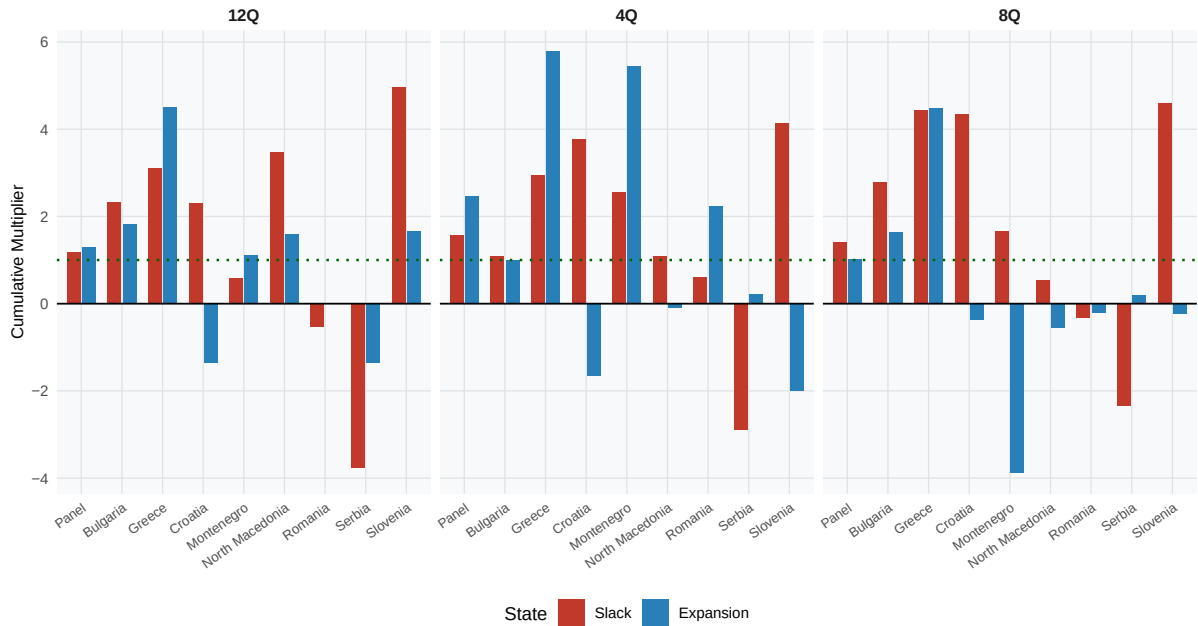
**Countries with moderate or state-invariant multipliers.** Bulgaria, Greece, Montenegro, and North Macedonia display multipliers that are broadly positive in both states without statistically significant cross-state differences. Greece is notable for large multipliers in both states at medium horizons ( $\hat{\beta}_{\text{slack}} = 4.43$ ,  $\hat{\beta}_{\text{exp}} = 4.49$  at  $h = 8$ ), likely reflecting fiscal transmission through the sovereign debt channel during the 2010–2018 crisis period. Bulgaria shows slack multipliers rising with the horizon (from 1.08 to 2.32), while Montenegro’s limited sample ( $T = 40$ ) implies estimates should be treated with caution.

**Countries with negative or subdued multipliers.** Romania and Serbia stand out with notably weak or negative multipliers. In Romania the slack multiplier turns negative at horizons 8 and 12 ( $-0.33$  and  $-0.54$  respectively), consistent with a crowding-out interpretation or with the view that Romanian fiscal expansions have been associated with debt-financed current transfers rather than productive public investment. Serbia displays uniformly negative slack multipliers across all horizons, reaching  $-3.77$  at  $h = 12$ . This striking result may partly reflect the short sample window (2010:Q3–2026:Q1) and Serbia’s post-2012 fiscal consolidation programme, during which spending cuts were accompanied by output compression.

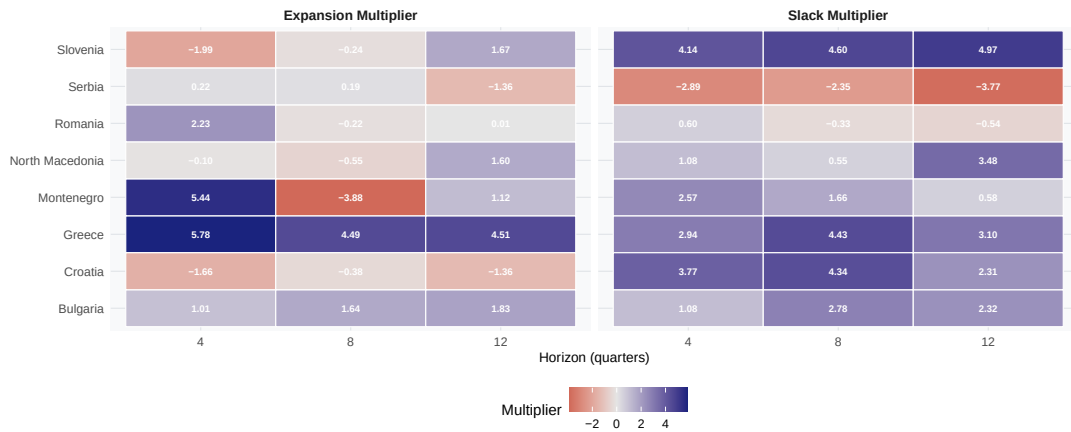


**Figure 5: Cumulative Government Spending Multipliers**

Green dotted line = multiplier of 1.0 (spending pays for itself)  
 Ramey-Zubairy (2018) IV scaling

**Figure 6: Cumulative Government Spending Multipliers****Figure 7: Multiplier Heatmap – State and Country**

Cumulative GDP response per 1pp spending shock at selected horizons

**Figure 7: Cumulative Multiplier Heatmap Matrix Summary**

#### 4.4 Robustness: Alternative Slack Definitions

Table 7 reports cumulative multipliers under three alternative percentile-based slack definitions. “High unemployment” is defined as unemployment above the country-specific 25th ( $\mathcal{P}_{25}$ ), 50th ( $\mathcal{P}_{50}$ ), or 75th ( $\mathcal{P}_{75}$ ) percentile.

Table 7: Robustness: Cumulative Multipliers under Alternative Slack Definitions

| Definition             | $h = 4$ |       | $h = 8$ |       | $h = 12$ |       | $p_8, p_{12}$  |
|------------------------|---------|-------|---------|-------|----------|-------|----------------|
|                        | Slack   | Exp.  | Slack   | Exp.  | Slack    | Exp.  |                |
| HP trend (baseline)    | 1.573   | 2.472 | 1.405   | 1.026 | 1.190    | 1.296 | 0.472, 0.794   |
| $u > \mathcal{P}_{25}$ | 1.957   | 2.725 | 1.236   | 1.110 | 1.209    | 1.750 | 0.876, 0.319   |
| $u > \mathcal{P}_{50}$ | 2.174   | 2.183 | 1.628   | 0.900 | 1.974    | 0.924 | <0.001, <0.001 |
| $u > \mathcal{P}_{75}$ | 1.259   | 2.248 | 0.710   | 1.385 | 1.658    | 1.235 | 0.169, 0.423   |

Notes:  $p_8$  and  $p_{12}$  are  $p$ -values for the Wald test  $H_0: \hat{\beta}_{\text{slack}} = \hat{\beta}_{\text{exp}}$  at horizons  $h = 8$  and  $h = 12$ . All specifications include country fixed effects and country-clustered SE.

The  $\mathcal{P}_{25}$  and  $\mathcal{P}_{75}$  thresholds produce results qualitatively similar to the HP-trend baseline: multipliers are positive in both states and the Wald test fails to reject equality at conventional levels.

The median-based definition ( $\mathcal{P}_{50}$ ) is the notable exception. At horizons of 8 and 12 quarters the slack multiplier (1.63, 1.97) significantly exceeds the expansion multiplier (0.90, 0.92), with  $p$ -values below 0.001. Under this definition, which splits each country's sample into above- and below-median unemployment periods, the data do suggest meaningful state dependence, with medium-run multipliers approaching 2.0 during high-unemployment episodes. One reason the median-split may outperform the HP-trend baseline is that the HP filter's two-sided smoothing can distort the identification of cyclical turning points near sample endpoints—the well-known “end-point problem” highlighted by Hamilton (2018).

## 5 Discussion

Taken together, the results paint a nuanced picture of fiscal transmission in South-Eastern Europe. At the pooled panel level, we find no statistically significant difference between slack and expansion multipliers, a finding robust to the HP-trend and percentile-based ( $\mathcal{P}_{25}$ ,  $\mathcal{P}_{75}$ ) slack definitions. Both multipliers are nonetheless positive and economically meaningful at medium horizons: at  $h = 12$ ,  $\hat{\beta}_{\text{slack}} = 1.19$  and  $\hat{\beta}_{\text{exp}} = 1.30$ , suggesting that government spending generates more than a unit return in cumulative output, on average, across the region.

Country-level heterogeneity is substantial and clusters along institutional lines. Slovenia and Croatia—the two euro-area members in the panel—display large, statistically sig-

nificant slack multipliers consistent with the Keynesian demand-slack hypothesis. One mechanism is monetary accommodation: euro-area membership eliminates exchange-rate appreciation as an offset to fiscal stimulus, and the ECB’s role as lender of last resort may reduce interest-rate crowding-out during recessions. Romania and Serbia exhibit multipliers below unity or negative in the slack state, pointing to structural differences in fiscal effectiveness—including spending composition, institutional quality, and fiscal consolidation pressures—that warrant further research.

The robustness analysis highlights the sensitivity of state-dependent conclusions to the measurement of economic slack. The  $\mathcal{P}_{50}$  threshold yields significant state dependence at medium horizons while the HP-trend baseline does not. This sensitivity echoes the concerns raised by Ramey and Zubairy (2018) and Hamilton (2018) regarding the definition of economic slack, and underscores the importance of reporting results under multiple definitions.

## 6 Conclusion

This paper estimates state-dependent government spending multipliers for eight South-Eastern European economies using quarterly Eurostat data and the LP-IV methodology of Owyang et al. (2013) with Ramey and Zubairy (2018) scaling. Our main conclusions are threefold.

First, at the pooled panel level, there is no statistically significant difference between fiscal multipliers in slack and expansion states across any horizon up to 12 quarters. Both multipliers exceed unity at medium horizons, suggesting broadly effective fiscal policy in the region.

Second, country-level results are highly heterogeneous. Slovenia and Croatia show large, statistically significant slack multipliers (up to 5.0 and 4.3) consistent with the Keynesian state-dependence hypothesis; Romania and Serbia exhibit subdued or negative slack multipliers, pointing to structural constraints on fiscal transmission. This heterogeneity aligns with the countries’ institutional positions: euro-area members benefit from monetary accommodation that amplifies fiscal demand effects.

Third, state-dependent conclusions are sensitive to the measurement of the business-cycle state. A median-split definition of slack ( $\mathcal{P}_{50}$ ) yields significant state dependence ( $p < 0.001$ ) at 8- and 12-quarter horizons while the HP-trend baseline does not. This reinforces calls in the literature for transparency and robustness analysis in the measurement of

slack.

Future work could extend this analysis by incorporating the composition of government spending (investment vs. transfers), by accounting for fiscal rules and debt sustainability constraints, and by applying heterogeneous-panel methods that allow slope coefficients to vary continuously across countries.

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